Seeneyu Database Schema Reference

> *Prepared for: Investor Tech Audit Workshop Date: 2026-03-29 | Version: 1.0 Database: PostgreSQL (Neon) | ORM: Prisma 6 | Schema: 915 lines*

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1. Schema Summary

Metric	Value
Total Models	45
Schema Lines	915
Total Indexes	62
Unique Constraints	14
Relations	38
Domain Groups	8

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2. Models by Domain

2.1 Auth & Users (5 models)

Model	Purpose	Fields	Key Fields	Indexes
User	User account with role and approval status	19	`email` (unique), `role`, `status`, `plan`, `passwordHash`, `onboardingComplete`	`status`
Account	OAuth provider accounts (NextAuth)	12	`userId`, `provider`, `providerAccountId`	`[provider, providerAccountId]` (unique)
AuthSession	Active login sessions	4	`sessionToken` (unique), `userId`, `expires`	-
VerificationToken	Email verification tokens	3	`identifier`, `token` (unique)	`[identifier, token]` (unique)
SkillBaseline	Onboarding skill self-assessment	6	`userId`, `skillCategory`, `level`, `selfRating`	`userId`, `[userId, skillCategory]` (unique)

Relationships:

- User has many: Account, AuthSession, UserSession, SkillBaseline, FoundationProgress, ArcadeAttempt, ActivityEvent, Subscription, Comment, AssistantConversation
- User has one: UserGamification, LearnerProfile

2.2 Core Learning Content (5 models)

Model	Purpose	Fields	Key Fields	Indexes
Clip	Learning content unit (movie clip)	22	`youtubeVideoId`, `skillCategory`, `difficulty`, `difficultyScore`, `script`, `observationGuide` (JSON), `mediaType`, `mediaUrl`	`skillCategory`, `difficulty`, `[skillCategory, difficulty]`
Annotation	Timed clip annotations	5	`clipId`, `atSecond`, `note`, `type`	`clipId`, `[clipId, atSecond]`
PracticeStep	Micro-practice step definitions	7	`clipId`, `stepNumber`, `skillFocus`, `instruction`, `targetDurationSec`	`clipId`
UserSession	Full performance recording session	12	`clipId`, `userId`, `recordingUrl`, `frameUrls`, `feedback` (JSON), `scores` (JSON)	`clipId`, `status`, `createdAt`
MicroSession	Micro-practice recording session	10	`clipId`, `stepNumber`, `recordingUrl`, `audioUrl`, `transcript`, `feedback` (JSON)	`clipId`

Relationships:

- Clip has many: Annotation, UserSession, PracticeStep
- UserSession belongs to: Clip, User

2.3 Foundation Education (5 models)

Model	Purpose	Fields	Key Fields	Indexes
FoundationCourse	Course catalog (3 courses)	7	`slug` (unique), `title`, `icon`, `color`, `order`	-
FoundationLesson	Lesson content	6	`courseId`, `slug`, `title`, `theoryHtml` (Text), `order`	`courseId`, `[courseId, slug]` (unique)
LessonExample	YouTube examples per lesson	7	`lessonId`, `youtubeId`, `title`, `mediaUrl`, `mediaType`	`lessonId`
QuizQuestion	Quiz questions with MCQ	7	`lessonId`, `question`, `options` (JSON), `correctIndex`, `explanation`	`lessonId`
FoundationProgress	Per-user lesson completion	8	`userId`, `lessonId`, `quizScore`, `quizPassed`, `completedAt`	`userId`, `[userId, lessonId]` (unique)

Relationships:

- FoundationCourse has many: FoundationLesson
- FoundationLesson has many: LessonExample, QuizQuestion, FoundationProgress, Comment

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2.4 Arcade (3 models)

Model	Purpose	Fields	Key Fields	Indexes
ArcadeBundle	Challenge collection (theme)	7	`title`, `theme`, `difficulty`, `xpReward`	-
ArcadeChallenge	Individual 10s challenge	12	`bundleId`, `type` (facial/gesture), `referenceImageUrl`, `mediaUrl`, `mediaType`, `orderIndex`	`bundleId`
ArcadeAttempt	User attempt record	8	`userId`, `challengeId`, `score`, `breakdown` (JSON), `feedbackLine`	`userId`, `challengeId`, `[userId, challengeId]`

Relationships:

- ArcadeBundle has many: ArcadeChallenge
- ArcadeChallenge has many: ArcadeAttempt, Comment
- ArcadeAttempt belongs to: User, ArcadeChallenge

2.5 Gamification (5 models)

Model	Purpose	Fields	Key Fields	Indexes
UserGamification	Per-user gamification stats	10	`userId` (unique), `totalXp`, `level`, `currentStreak`, `longestStreak`, `hearts`, `heartsRefillAt`, `streakFreezes`	`totalXp`, `currentStreak`
XpTransaction	XP award history	6	`userId`, `amount`, `source`, `sourceId`, `metadata` (JSON)	`[userId, createdAt]`
DailyQuest	Daily challenge goals	9	`userId`, `date`, `questType`, `target`, `progress`, `xpReward`, `completed`	`[userId, date]`, `[userId, date, questType]` (unique)
Badge	Achievement definitions	7	`slug` (unique), `name`, `category`, `criteria` (JSON), `iconEmoji`	-
UserBadge	Earned badge records	4	`userId`, `badgeId`, `earnedAt`	`userId`, `[userId, badgeId]` (unique)

Relationships:

- UserGamification belongs to: User (1:1)
- Badge has many: UserBadge

2.6 Social & Communication (6 models)

Model	Purpose	Fields	Key Fields	Indexes

Comment	User comments (1-level threading)	13	`userId`, `lessonId`, `challengeId`, `parentId`, `body` (Text), `isHidden`	`[lessonId, createdAt]`, `[challengeId, createdAt]`, `parentId`, `userId`
AssistantConversation	Coach Ney AI chat session	5	`userId`, `context`	`userId`, `[userId, context]`
AssistantMessage	Chat messages (user + assistant)	5	`conversationId`, `role`, `content` (Text), `audioUrl`	`[conversationId, createdAt]`
Leaderboard	Cached leaderboard snapshots	5	`type`, `period`, `entries` (JSON)	`[type, period]` (unique)
UserFollow	Social follow relationships	4	`followerId`, `followingId`	`followerId`, `followingId`, `[followerId, followingId]` (unique)
PushSubscription	Web push subscription endpoints	4	`userId`, `endpoint` (Text), `keys` (JSON)	`userId`

2.7 Content Management & Admin (9 models)

Model	Purpose	Fields	Key Fields	Indexes
CrawlJob	YouTube content discovery job	10	`skillCategory`, `keywords` (JSON), `status`, `createdBy`	`status`, `createdBy`
CrawlResult	Crawl candidate videos	12	`jobId`, `youtubeId`, `relevanceScore`, `aiAnalysis`, `status`	`jobId`, `[jobId, status]`
CmsPage	CMS-managed pages	8	`slug` (unique), `title`, `content` (JSON), `status`	`[slug, status]`
BlogPost	Blog posts	10	`slug` (unique), `title`, `body` (Text), `coverImage`, `status`	`[status, publishedAt]`
SiteSettings	Key-value site settings	3	`key` (unique), `value` (JSON)	-
TeamMember	Team page members	7	`name`, `title`, `bio`, `avatarUrl`, `order`	`[isActive, order]`
ContentSource	Data crawler raw content	9	`type`, `url`, `rawContent` (Text), `status`	`[type, status]`, `createdAt`
ExpressionAsset	Expression reference images	8	`imageUrl`, `label`, `tags` (JSON), `confidence`, `status`	`label`, `status`
ErrorLog	Application error tracking	11	`level`, `source`, `message` (Text), `stack`, `resolved`	`level`, `source`, `resolved`, `createdAt`

2.8 Subscription & Payments (3 models)

Model	Purpose	Fields	Key Fields	Indexes
Plan	Subscription tier definitions	9	`slug` (unique), `name`, `monthlyPrice`, `annualPrice`, `features` (JSON), `videoLimitSec`	-
Subscription	Active user subscriptions	10	`userId`, `planId`, `status`, `period`, `startDate`, `endDate`	`userId`, `status`
Payment	Payment transaction records	9	`subscriptionId`, `gateway`, `gatewayOrderId`, `amount`, `currency`, `status`	`subscriptionId`, `[gateway, gatewayOrderId]`

2.9 Analytics & Data Pipeline (4 models)

Model	Purpose	Fields	Key Fields	Indexes
ActivityEvent	User activity tracking	5	`userId`, `type`, `metadata` (JSON)	`userId`, `type`, `createdAt`, `[userId, type]`
AnalysisMetric	MediaPipe performance metrics	7	`sessionType`, `durationMs`, `faceDetected`, `poseDetected`, `score`	`sessionType`, `createdAt`
GameDataExport	Data export jobs	8	`name`, `format`, `filters` (JSON), `fileUrl`, `status`	`status`
TrainingDataLabel	ML training data labels	6	`submissionId`, `expressionLabel`, `confidence`, `validatedBy`	`expressionLabel`, `submissionId`

2.10 Mini-Games & Data Collection (4 models)

Model	Purpose	Fields	Key Fields	Indexes
MiniGame	Game type definitions (5 games)	6	`type` (unique), `title`, `config` (JSON), `isActive`	-
MiniGameRound	Game round content	6	`gameId`, `orderIndex`, `prompt`, `imageUrl`, `correctAnswer`, `options` (JSON)	`gameId`
MiniGameSession	Play session records	8	`gameId`, `playerId`, `score`, `totalRounds`, `responses` (JSON)	`[gameId, score]`, `playerId`
ExpressionSubmission	Camera expression captures	7	`challengeLabel`, `imageUrl`, `aiScore`, `aiAnalysis`, `status`	`challengeLabel`, `status`

2.11 Learning Assistant Engine (5 models)

Model	Purpose	Fields	Key Fields	Indexes
LearnerProfile	Learner preferences and analytics	16	`userId` (unique), `timezone`, `preferredChannels` (JSON), `engagementScore`, `weakSkills` (JSON), `strongSkills` (JSON), `whatsappPhone`	`engagementScore`
LearningPlan	Daily/weekly activity plans	8	`userId`, `type`, `date`, `activities` (JSON), `status`	`[userId, type, date]` (unique), `[userId, status]`
ScheduledNotification	Notification delivery queue	11	`userId`, `triggerType`, `channel`, `scheduledFor`, `payload` (JSON), `priority`, `status`, `attempts`	`[status, scheduledFor]`, `[userId, triggerType]`
NotificationLog	Delivery audit trail	12	`userId`, `notificationId`, `triggerType`, `channel`, `title`, `deliveryStatus`, `deliveredAt`, `openedAt`	`[userId, createdAt]`, `triggerType`
NotificationTemplate	Message templates with variables	10	`slug` (unique), `triggerType`, `channel`, `title`, `body` (Text), `variables` (JSON), `locale`	`[triggerType, channel, locale]` (unique)

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3. Data Volume Estimates

3.1 Per-User Data Generation

Data Type	Records per Active User per Day	Size per Record
ActivityEvent	5-15	~200 bytes
XpTransaction	3-8	~150 bytes
DailyQuest	3	~200 bytes
FoundationProgress	0-2	~150 bytes
ArcadeAttempt	0-5	~300 bytes
MicroSession	0-3	~500 bytes
UserSession	0-1	~500 bytes
Comment	0-2	~300 bytes
ScheduledNotification	2-4	~400 bytes
NotificationLog	2-4	~300 bytes

AssistantMessage	0-10	~500 bytes

3.2 Projected Table Sizes

Scale	Users	DAU (30%)	Monthly Records	DB Size (est.)
1K users	1,000	300	~400K	~200 MB
10K users	10,000	3,000	~4M	~2 GB
100K users	100,000	30,000	~40M	~20 GB

3.3 Storage Volume (Vercel Blob)

Scale	Recording Uploads/Day	Storage/Month
1K users	~150 recordings	~15 GB
10K users	~1,500 recordings	~150 GB
100K users	~15,000 recordings	~1.5 TB

3.4 Growth-Critical Tables (Require Partition/Archive Strategy at 100K+)

Table	Growth Rate	Mitigation
ActivityEvent	~15 per user/day	Archive after 90 days
XpTransaction	~8 per user/day	Archive after 1 year
NotificationLog	~4 per user/day	Archive after 30 days
ScheduledNotification	~4 per user/day (cleared after send)	Auto-cleanup sent records
AssistantMessage	~10 per active chat session	Summarize + archive old conversations
MiniGameSession	Variable (public, no auth required)	TTL-based cleanup

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4. Index Strategy

Composite Indexes for Common Queries

```
-- User dashboard (recent activity)
@@index([userId, type])          -- ActivityEvent
@@index([userId, createdAt])     -- XpTransaction, NotificationLog

-- Admin analytics
@@index([createdAt])             -- ActivityEvent, AnalysisMetric
@@index([status])                -- CrawlJob, ErrorLog, Subscription

-- Learning engine queries
@@index([status, scheduledFor])  -- ScheduledNotification (queue processing)
@@index([userId, status])        -- LearningPlan (active plans)
```

```
@@index([engagementScore])       -- LearnerProfile (batch processing)

-- Content discovery
@@index([skillCategory, difficulty]) -- Clip (library filtering)
@@index([jobId, status])            -- CrawlResult (job review)
```

Missing Indexes (Technical Debt)

The current schema is well-indexed for known access patterns. Potential additions at scale:

- `ActivityEvent` partitioning by `createdAt` (time-series optimization)
- `UserSession.userId` index for per-user session history
- `MiniGameSession.createdAt` for time-based cleanup queries

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5. Entity-Relationship Summary

```
User ----1:N---- UserSession ----N:1---- Clip
  |                                        |
  +----1:N---- ArcadeAttempt --N:1-- ArcadeChallenge --N:1-- ArcadeBundle
  |
  +----1:N---- FoundationProgress --N:1-- FoundationLesson --N:1-- FoundationCourse
  |
  +----1:1---- UserGamification
  +----1:1---- LearnerProfile
  +----1:N---- Subscription ----N:1---- Plan
  |                |
  |                +----1:N---- Payment
  |
  +----1:N---- Comment
  +----1:N---- AssistantConversation ----1:N---- AssistantMessage
  +----1:N---- ActivityEvent
  +----1:N---- XpTransaction
  +----1:N---- DailyQuest
  +----1:N---- UserBadge ----N:1---- Badge
  +----1:N---- SkillBaseline
  +----1:N---- Account
```

Document prepared by Data Engineer role. Schema verified against `prisma/schema.prisma` (915 lines, 45 models) as of 2026-03-29.